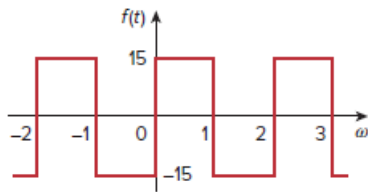


Problem

Find the Fourier series of the square wave in Fig. 17.5. Plot the amplitude and phase spectra.

Figure 17.5:



Step-by-step solution

Step 1 of 7

Refer to square wave in Figure 17.5 in the textbook.

The function is,

$$f(t) = \begin{cases} 15 & 0 < t < 1 \\ -15 & 1 < t < 2 \end{cases}$$

Here,

$$T = 2$$

The fundamental frequency is,

$$\begin{aligned} \omega_0 &= \frac{2\pi}{T} \\ &= \frac{2\pi}{2} \\ &= \pi \end{aligned}$$

Since the function is odd

The value of a_n is 0.

$$a_n = 0$$

The Fourier series is,

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Step 2 of 7

Calculate the value of b_n .

$$\begin{aligned}
 b_n &= \frac{2}{T} \int_0^T f(t) \sin(n\omega_0 t) dt \\
 &= \frac{2}{2} \int_0^4 f(t) \sin(n\omega_0 t) dt \\
 &= \int_0^1 (15) \sin(n\pi t) dt + \int_1^2 (-15) \sin(n\pi t) dt \\
 &= (15) \left[-\frac{\cos(n\pi t)}{n\pi} \right]_{t=0}^{t=1} - (15) \left[-\frac{\cos(n\pi t)}{n\pi} \right]_{t=1}^{t=2} \\
 &= (15) \left(\frac{1 - \cos(n\pi)}{n\pi} \right) + (15) \left(\frac{1 - \cos(n\pi)}{n\pi} \right) \\
 &= (30) \left(\frac{1 - \cos(n\pi)}{n\pi} \right) \\
 &= \frac{60}{n\pi} \quad n \text{ is odd}
 \end{aligned}$$

Step 3 of 7

Calculate the Fourier series for the sine wave.

$$\begin{aligned}
 f(t) &= a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) \\
 &= 0 + \sum_{n=1}^{\infty} \left[(0) \cos n\omega_0 t + \left(\frac{60}{n\pi} \right) \sin n\omega_0 t \right] \\
 &= \frac{60}{\pi} \sum_{\substack{n=1 \\ n=\text{odd}}}^{\infty} \left\{ \frac{1}{n} \sin(n\pi t) \right\}
 \end{aligned}$$

Therefore, the Fourier series, $f(t)$ of the periodic function is

$$\boxed{\frac{60}{\pi} \sum_{k=1}^{\infty} \left\{ \frac{1}{n} \sin(n\pi t) \right\}, n = 2k - 1}$$

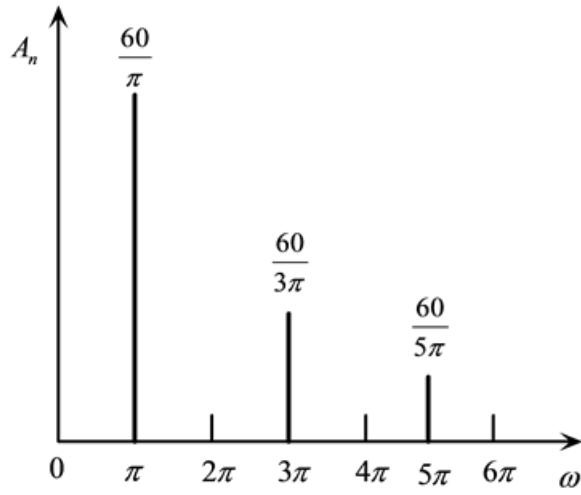
Step 4 of 7

Amplitude of n -th harmonic is,

$$\begin{aligned}
 A_n &= \sqrt{a_n^2 + b_n^2}, \quad n \text{ is odd} \\
 &= b_n, \quad n \text{ is odd} \\
 &= \frac{60}{n\pi}, \quad n \text{ is odd}
 \end{aligned}$$

Step 5 of 7

The amplitude spectrum is shown in Figure 1.



Therefore, the amplitude spectrum is shown in Figure 1.

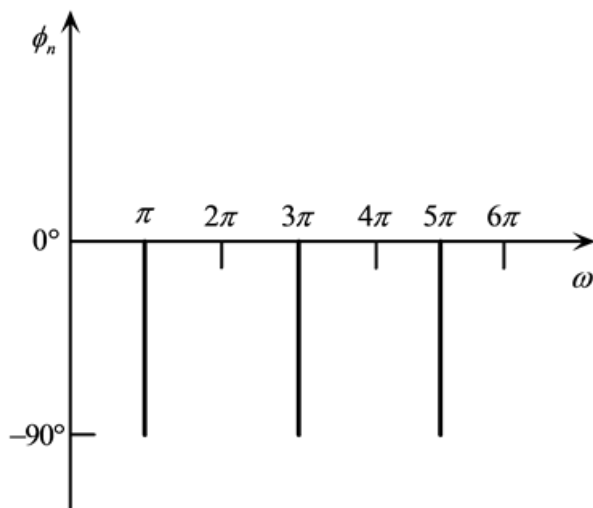
Step 6 of 7

The phase ϕ_n is,

$$\begin{aligned}\phi_n &= -\tan^{-1}\left(\frac{b_n}{a_n}\right), n \text{ is odd} \\ &= -\tan^{-1}\left(\frac{b_n}{0}\right), n \text{ is odd} \\ &= -\tan^{-1}(\infty), n \text{ is odd} \\ &= -90^\circ, n \text{ is odd}\end{aligned}$$

Step 7 of 7

The phase spectrum is shown in Figure 2.



Therefore, the phase spectrum is shown in Figure 2.